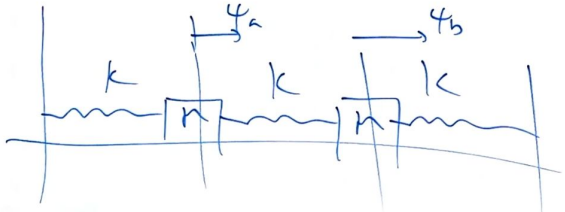
28.7.16

Wave Lec 22

Coupled oscillations : two-mass spring system



$$M\ddot{\psi}_a = -k\psi_a + k(\psi_b - \psi_a)$$

$$M\ddot{\psi}_b = -k\psi_b - k(\psi_b - \psi_a)$$

Two normal modes

Mode 1) $\psi_a = \psi_b$, $\omega_1^2 = k/M$

Mode 2) $\psi_b = -\psi_a$, $\omega_2^2 = 3k/M$

(1a) $\ddot{\psi}_a = -\frac{2k}{M}\psi_a + \frac{k}{M}\psi_b$

(1b) $\ddot{\psi}_b = +\frac{k}{M}\psi_a - \frac{2k}{M}\psi_b$

(2a) $\ddot{\psi}_a = -\alpha\psi_a - \beta\psi_b$

(2b) $\ddot{\psi}_b = -\beta\psi_a - \alpha\psi_b$

(3) $\alpha = \frac{2k}{M}$, $\beta = -k/M$

Normal mode Analysis:

(4a) $\psi_a = A\cos(\omega t + \varphi)$

(4b) $\psi_b = B\cos(\omega t + \varphi)$

Put $(4a, 4b)$ into $(2a, 2b)$

$$-\omega^2\psi_a = -\alpha\psi_a - \beta\psi_b$$

$$-\omega^2\psi_b = -\beta\psi_a - \alpha\psi_b$$

(5a) $(\alpha - \omega^2)\psi_a + \beta\psi_b = 0$

(5b) $\beta\psi_a + (\alpha - \omega^2)\psi_b = 0$

(5a) $\Rightarrow$

$$\frac{\psi_b}{\psi_a} = \frac{\omega^2 - \alpha}{\beta} \quad (6a)$$

(5b) $\Rightarrow$ $\frac{\psi_b}{\psi_a} = \frac{\beta}{\omega^2 - \alpha}$ (6b)

(6) $\Rightarrow$ $\frac{\omega^2 - \alpha}{\beta} = \frac{\beta}{\omega^2 - \alpha}$ (7)

(7) $\Rightarrow$

(8)

(9)

(11)

(11) $\left(\frac{\psi_b}{\psi_a}\right)_{mode}$

(12)

$-\beta\psi_b$

$-\alpha\psi_b$

$= 0$

$\omega^2)\psi_b = 0$

(6a)

(6b)

(7) $\Rightarrow$ $(\omega^2-\alpha)^2 = \beta^2$

$\omega^2 - \alpha = \pm\sqrt{\beta^2}$

(8) $\omega^2 = \alpha \pm |\beta|$

$\omega_1^2 = \alpha - |\beta|$

(9) $= \frac{2k}{m} - \frac{k}{m} = \frac{k}{m}$

(11) $\omega_2^2 = \alpha + |\beta|$

$= \frac{2k}{m} + \frac{k}{m} = \frac{3k}{m}$

(11) $\left(\frac{\psi_b}{\psi_a}\right)_{mode1} = \left(\frac{B\cos(\omega t+\varphi)}{A\cos(\omega t+\varphi)}\right)_{mode1} = \frac{B_1}{A_1} = \left(\frac{\omega^2-\alpha}{\beta}\right)_1$

(12) $B_1/A_1 = \frac{\omega_1^2-\alpha}{\beta} = \frac{k/m - 2k/m}{-k/m} = 1$

(13) $\left(\frac{\psi_b}{\psi_a}\right)_{mode2} = \left(\frac{B\cos(\omega t+\varphi)}{A\cos(\omega t+\varphi)}\right)_{mode2} = \frac{B_2}{A_2}$

$= \left(\frac{\omega^2-\alpha}{\beta}\right)_{mode2} = \frac{\omega_2^2-\alpha}{\beta} \Rightarrow$

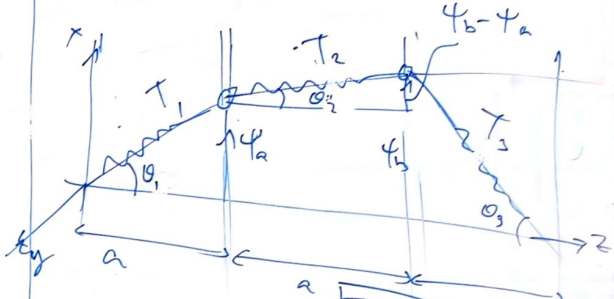
(14) $\frac{B_2}{A_2} = \frac{3k/m - 2k/m}{-k/m} = -1$

$B_1 = -A_1, \quad B_2 = A_2$

(15a) $\psi_a = A_1\cos(\omega_1 t+\varphi_1) + A_2\cos(\omega_2 t+\varphi_2)$

(15b) $\psi_b = A_1\cos(\omega_1 t+\varphi_1) - A_2\cos(\omega_2 t+\varphi_2)$

Transverse coupled DOF of two-mass-spring sys



$T_1 \approx T_2 \approx T_3 = T_0$

$\ddot{\psi}_a = -T_1\sin\theta_1 + T_2\sin\theta_2$

$\approx -T_1\theta_1 + T_2\theta_2$

$\approx -T_1\frac{\psi_a}{a} + T_2\frac{\psi_b-\psi_a}{a}$

$\ddot{\psi}_a \approx -\frac{T_0}{m_a}\psi_a + \frac{T_0}{m_a}(\psi_b-\psi_a)$